Derivation of Jacobian matrix

Given n+1 components in a melt with their concentrations w_1, …w_n and w_{n+1}, let's choose the n+1th component as the dependent component and derive their derivatives with respect to eigenvectors and eigenvalues as well as interface position.

Define w as follows:

$$\boldsymbol{w} = (w_1,\ w_2, \ldots,\ w_n)^T \tag{1}$$

where

$$w_{n+1} = 1 - \sum_{i=1}^{n} w_i \tag{2}$$

Here we assume in both halves of glasses, the concentration is not necessarily homogeneous, but the concentration changes linearly with distance to the interface respectively. In the left part, assume

$$\boldsymbol{w}(x, t = 0) = \boldsymbol{w_1} + \boldsymbol{k_1}(x - x_0)$$

and in the right part assume

$$\boldsymbol{w}(x, t = 0) = \boldsymbol{w_2} + \boldsymbol{k_2}(x - x_0)$$

For diffusion couple:

$$\boldsymbol{w} = \overline{\boldsymbol{w}} + \overline{\boldsymbol{k}}\,(x - x_0) + P \times E \times P^{-1} \times \left(\frac{\Delta \boldsymbol{w}}{2} + \frac{\Delta \boldsymbol{k}}{2}(x - x_0)\right)$$

$$+P \times F \times P^{-1} \times \Delta \boldsymbol{k} \tag{3}$$

where

$$\overline{\boldsymbol{w}} = \frac{\boldsymbol{w_2} + \boldsymbol{w_1}}{2} \tag{4}$$

$$\overline{\boldsymbol{k}} = \frac{\boldsymbol{k_2} + \boldsymbol{k_1}}{2} \tag{5}$$

$$\Delta \boldsymbol{w} = \boldsymbol{w_2} - \boldsymbol{w_1} \tag{6}$$

$$\Delta \boldsymbol{k} = \boldsymbol{k_2} - \boldsymbol{k_1} \tag{7}$$

$$E = \begin{pmatrix} \operatorname{erf}\left(\frac{x - x_0}{\sqrt{4e^{\beta_1}t}}\right) & & \\ & \ddots & \\ & & \operatorname{erf}\left(\frac{x - x_0}{\sqrt{4e^{\beta_n}t}}\right) \end{pmatrix} = \begin{pmatrix} \operatorname{erf}\left(\frac{x - x_0}{\sqrt{4t}} e^{-\frac{\beta_1}{2}}\right) & & \\ & \ddots & \\ & & \operatorname{erf}\left(\frac{x - x_0}{\sqrt{4t}} e^{-\frac{\beta_n}{2}}\right) \end{pmatrix} \tag{8}$$

and

$$F=\begin{pmatrix}\sqrt{\frac{e^{\beta_1 t}}{\pi}}\mathrm{e}^{-\frac{(x-x_0)^2}{4e^{\beta_1 t}}} & & \\ & \ddots & \\ & & \sqrt{\frac{e^{\beta_n t}}{\pi}}\mathrm{e}^{-\frac{(x-x_0)^2}{4e^{\beta_n t}}}\end{pmatrix}=\begin{pmatrix}\sqrt{\frac{t}{\pi}}\mathrm{e}^{-\frac{(x-x_0)^2}{4e^{\beta_1 t}}+\frac{\beta_1}{2}} & & \\ & \ddots & \\ & & \sqrt{\frac{t}{\pi}}\mathrm{e}^{-\frac{(x-x_0)^2}{4e^{\beta_n t}}+\frac{\beta_n}{2}}\end{pmatrix} \tag{9}$$

Define $\boldsymbol{\beta}$ as follows:

$$\boldsymbol{\beta}=\begin{pmatrix}\beta_1\\ \vdots\\ \beta_n\end{pmatrix} \tag{10}$$

Define $\boldsymbol{Y}$ as follows:

$$\boldsymbol{Y}=\begin{pmatrix}\frac{x-x_0}{\sqrt{4t}}e^{-\frac{\beta_1}{2}}\\ \vdots\\ \frac{x-x_0}{\sqrt{4t}}e^{-\frac{\beta_n}{2}}\end{pmatrix}=\frac{x-x_0}{\sqrt{4t}}e^{-\frac{\boldsymbol{\beta}}{2}} \tag{11}$$

Then

$$E=diag\begin{pmatrix}\operatorname{erf}\left(\frac{x-x_0}{\sqrt{4t}}e^{-\frac{\beta_1}{2}}\right)\\ \vdots\\ \operatorname{erf}\left(\frac{x-x_0}{\sqrt{4t}}e^{-\frac{\beta_n}{2}}\right)\end{pmatrix}=diag\big(\operatorname{erf}(\boldsymbol{Y})\big) \tag{12}$$

and

$$F=diag\begin{pmatrix}\sqrt{\frac{t}{\pi}}\mathrm{e}^{-\frac{(x-x_0)^2}{4e^{\beta_1 t}}+\frac{\beta_1}{2}}\\ \vdots\\ \sqrt{\frac{t}{\pi}}\mathrm{e}^{-\frac{(x-x_0)^2}{4e^{\beta_n t}}+\frac{\beta_n}{2}}\end{pmatrix}=\sqrt{\frac{t}{\pi}}diag\left(\exp\left(-\boldsymbol{Y}\odot\boldsymbol{Y}+\frac{1}{2}\boldsymbol{\beta}\right)\right) \tag{13}$$

According to Eq. (7),

$$\begin{aligned}\mathbf{d}\,E&=\mathbf{d}\,diag(\operatorname{erf}(\boldsymbol{Y}))\\ &=diag(\mathbf{d}\operatorname{erf}(\boldsymbol{Y}))\\ &=diag\left(\frac{2}{\sqrt{\pi}}e^{-\boldsymbol{Y}\odot\boldsymbol{Y}}\odot\mathbf{d}\,\boldsymbol{Y}\right)\\ &=diag\left(\frac{2}{\sqrt{\pi}}e^{-\boldsymbol{Y}\odot\boldsymbol{Y}}\odot\mathbf{d}\left(\frac{x-x_0}{\sqrt{4t}}e^{-\frac{\boldsymbol{\beta}}{2}}\right)\right)\\ &=diag\left(\frac{2}{\sqrt{\pi}}e^{-\boldsymbol{Y}\odot\boldsymbol{Y}}\odot\left(\frac{x-x_0}{\sqrt{4t}}*\left(-\frac{1}{2}\right)*e^{-\frac{\boldsymbol{\beta}}{2}}\odot\mathbf{d}\,\boldsymbol{\beta}-\frac{1}{\sqrt{4t}}\,e^{-\frac{\boldsymbol{\beta}}{2}}\mathbf{d}x_0\right)\right)\end{aligned}$$

$$= -\frac{x-x_0}{\sqrt{4\pi t}} diag\left(e^{-\boldsymbol{Y}\odot\boldsymbol{Y}} \odot e^{-\frac{\boldsymbol{\beta}}{2}} \odot \mathbf{d}\,\boldsymbol{\beta}\right) - \frac{1}{\sqrt{\pi t}} diag\left(e^{-\boldsymbol{Y}\odot\boldsymbol{Y}} \odot e^{-\frac{\boldsymbol{\beta}}{2}}\right) \mathbf{d}\, x_0$$

$$= -\frac{x-x_0}{\sqrt{4\pi t}} diag\left(e^{-\boldsymbol{Y}\odot\boldsymbol{Y}} \odot e^{-\frac{\boldsymbol{\beta}}{2}}\right) \odot \mathbf{d}\,(diag\,\boldsymbol{\beta}) - \frac{1}{\sqrt{\pi t}} diag\left(e^{-\boldsymbol{Y}\odot\boldsymbol{Y}} \odot e^{-\frac{\boldsymbol{\beta}}{2}}\right) \mathbf{d}\, x_0$$

$$= dE_1 \mathbf{d}\, x_0 + dE_2 \odot \mathbf{d}\,(diag\,\boldsymbol{\beta}) \tag{14}$$

where

$$dE_1 = -\frac{1}{\sqrt{\pi t}} diag\left(e^{-\boldsymbol{Y}\odot\boldsymbol{Y}} \odot e^{-\frac{\boldsymbol{\beta}}{2}}\right) \tag{15}$$

$$dE_2 = -\frac{x-x_0}{\sqrt{4\pi t}} diag\left(e^{-\boldsymbol{Y}\odot\boldsymbol{Y}} \odot e^{-\frac{\boldsymbol{\beta}}{2}}\right) \tag{16}$$

And we have

$$\mathbf{d}\, F = \sqrt{\frac{t}{\pi}} \mathbf{d}\, diag\left(\exp\left(-\boldsymbol{Y}\odot\boldsymbol{Y} + \frac{1}{2}\boldsymbol{\beta}\right)\right)$$

$$= \sqrt{\frac{t}{\pi}}\, diag\left(\mathbf{d}\exp\left(-\boldsymbol{Y}\odot\boldsymbol{Y} + \frac{1}{2}\boldsymbol{\beta}\right)\right)$$

$$= \sqrt{\frac{t}{\pi}}\, diag\left(\exp\left(-\boldsymbol{Y}\odot\boldsymbol{Y} + \frac{1}{2}\boldsymbol{\beta}\right) \odot \left(-2\boldsymbol{Y}\odot\mathbf{d}\,\boldsymbol{Y} + \frac{1}{2}\mathbf{d}\,\boldsymbol{\beta}\right)\right)$$

$$= \sqrt{\frac{t}{\pi}}\, diag\left(\exp\left(-\boldsymbol{Y}\odot\boldsymbol{Y} + \frac{1}{2}\boldsymbol{\beta}\right)\right) \odot diag\left(-2\boldsymbol{Y}\odot\mathbf{d}\,\boldsymbol{Y} + \frac{1}{2}\mathbf{d}\,\boldsymbol{\beta}\right)$$

$$= \sqrt{\frac{t}{\pi}}\, diag\left(\exp\left(-\boldsymbol{Y}\odot\boldsymbol{Y} + \frac{1}{2}\boldsymbol{\beta}\right)\right) \odot diag\left(\frac{1}{\sqrt{t}}\boldsymbol{Y}\odot e^{-\frac{\boldsymbol{\beta}}{2}}\mathbf{d}x_0 + \frac{x-x_0}{\sqrt{4t}}\boldsymbol{Y}\odot e^{-\frac{\boldsymbol{\beta}}{2}}\odot\mathbf{d}\,\boldsymbol{\beta} + \frac{1}{2}\mathbf{d}\,\boldsymbol{\beta}\right)$$

$$= \frac{x-x_0}{\sqrt{4\pi t}}\, diag\left(\exp\left(-\boldsymbol{Y}\odot\boldsymbol{Y} + \frac{1}{2}\boldsymbol{\beta}\right) \odot e^{-\boldsymbol{\beta}}\right)\mathbf{d}x_0$$

$$+ \sqrt{\frac{t}{\pi}}\, diag\left(\exp\left(-\boldsymbol{Y}\odot\boldsymbol{Y} + \frac{1}{2}\boldsymbol{\beta}\right)\right) \odot \left(diag\left(\frac{x-x_0}{\sqrt{4t}}\boldsymbol{Y}\odot e^{-\frac{\boldsymbol{\beta}}{2}}\right) + \frac{1}{2}I_n\right)\odot\mathbf{d}\,(diag\,\boldsymbol{\beta})$$

$$= dF_1 \mathbf{d}\, x_0 + dF_2 \odot \mathbf{d}\,(diag\,\boldsymbol{\beta}) \tag{17}$$

Where

$$dF_1 = \frac{x-x_0}{\sqrt{4\pi t}}\, diag\left(\exp\left(-\boldsymbol{Y}\odot\boldsymbol{Y} - \frac{1}{2}\boldsymbol{\beta}\right)\right) = -dE_2 \tag{18}$$

$$dF_2 = \sqrt{\frac{t}{\pi}}\, diag\left(\exp\left(-\boldsymbol{Y}\odot\boldsymbol{Y} + \frac{1}{2}\boldsymbol{\beta}\right)\right) \odot \left(diag\left(\frac{x-x_0}{\sqrt{4t}}\boldsymbol{Y}\odot e^{-\frac{\boldsymbol{\beta}}{2}}\right) + \frac{1}{2}I_n\right) \tag{19}$$

Take the total differential of $\boldsymbol{w}$:

$$\mathbf{d}\,\boldsymbol{w} = \mathbf{d}\,\overline{\boldsymbol{w}} + \mathbf{d}\,\overline{\boldsymbol{k}}\,(x-x_0) - \overline{\boldsymbol{k}}\,\mathbf{d}\,x_0$$

$$
\begin{aligned}
&+\mathbf{d}\,P \times E \times P^{-1} \times \left(\frac{\Delta \boldsymbol{w}}{2} + \frac{\Delta \boldsymbol{k}}{2}(x - x_0)\right) + P \times \mathbf{d}\,E \times P^{-1} \times \left(\frac{\Delta \boldsymbol{w}}{2} + \frac{\Delta \boldsymbol{k}}{2}(x - x_0)\right) \\
&+ P \times E \times \mathbf{d}\,P^{-1} \times \left(\frac{\Delta \boldsymbol{w}}{2} + \frac{\Delta \boldsymbol{k}}{2}(x - x_0)\right) \\
&+P \times E \times P^{-1} \times \left(\frac{\mathbf{d}\,\Delta \boldsymbol{w}}{2} + \frac{\mathbf{d}\,\Delta \boldsymbol{k}}{2}(x - x_0) - \frac{\Delta \boldsymbol{k}}{2}\mathbf{d}\,x_0\right) \\
&+\mathbf{d}P \times F \times P^{-1} \times \Delta \boldsymbol{k} + P \times \mathbf{d}F \times P^{-1} \times \Delta \boldsymbol{k} + P \times F \times \mathbf{d}P^{-1} \times \Delta \boldsymbol{k} \\
&+ P \times F \times P^{-1} \times \mathbf{d}\Delta \boldsymbol{k} \\
&{\color{red}=} \mathbf{d}\,\overline{\boldsymbol{w}} + \mathbf{d}\,\overline{\boldsymbol{k}}\,(x - x_0) - \overline{\boldsymbol{k}}\;\mathbf{d}\,x_0 \\
&+\mathbf{d}\,P \times E \times P^{-1} \times \left(\frac{\Delta \boldsymbol{w}}{2} + \frac{\Delta \boldsymbol{k}}{2}(x - x_0)\right) \\
&+ P \times \left(dE_1 \mathbf{d}\,x_0 + dE_2 \odot \mathbf{d}\,(diag\,\boldsymbol{\beta})\right) \times P^{-1} \times \left(\frac{\Delta \boldsymbol{w}}{2} + \frac{\Delta \boldsymbol{k}}{2}(x - x_0)\right) \\
&+ P \times E \times (-P^{-1} \times \mathbf{d}\,P \times P^{-1}) \times \left(\frac{\Delta \boldsymbol{w}}{2} + \frac{\Delta \boldsymbol{k}}{2}(x - x_0)\right) \\
&+P \times E \times P^{-1} \times \left(\frac{\mathbf{d}\,\Delta \boldsymbol{w}}{2} + \frac{\mathbf{d}\,\Delta \boldsymbol{k}}{2}(x - x_0) - \frac{\Delta \boldsymbol{k}}{2}\mathbf{d}\,x_0\right) \\
&+\mathbf{d}P \times F \times P^{-1} \times \Delta \boldsymbol{k} + P \times \left(dF_1 \mathbf{d}\,x_0 + dF_2 \odot \mathbf{d}\,(diag\,\boldsymbol{\beta})\right) \times P^{-1} \times \Delta \boldsymbol{k} \\
&+ P \times F \times (-P^{-1} \times \mathbf{d}\,P \times P^{-1}) \times \Delta \boldsymbol{k} \\
&+ P \times F \times P^{-1} \times \mathbf{d}\Delta \boldsymbol{k} \\
&{\color{red}=} \mathbf{d}\,\overline{\boldsymbol{w}} + \mathbf{d}\,\overline{\boldsymbol{k}}(x - x_0) - \overline{\boldsymbol{k}}\;\mathbf{d}\,x_0 \\
&+\mathbf{d}\,P \times E \times P^{-1} \times \left(\frac{\Delta \boldsymbol{w}}{2} + \frac{\Delta \boldsymbol{k}}{2}(x - x_0)\right) + P \times dE_1 \times P^{-1} \times \left(\frac{\Delta \boldsymbol{w}}{2} + \frac{\Delta \boldsymbol{k}}{2}(x - x_0)\right)\mathbf{d}\,x_0 \\
&+ P \times \left(dE_2 \odot \mathbf{d}\,(diag\,\boldsymbol{\beta})\right) \times P^{-1} \times \left(\frac{\Delta \boldsymbol{w}}{2} + \frac{\Delta \boldsymbol{k}}{2}(x - x_0)\right) \\
&- P \times E \times P^{-1} \times \mathbf{d}\,P \times P^{-1} \times \left(\frac{\Delta \boldsymbol{w}}{2} + \frac{\Delta \boldsymbol{k}}{2}(x - x_0)\right) \\
&+P \times E \times P^{-1} \times \left(\frac{\mathbf{d}\,\Delta \boldsymbol{w}}{2} + \frac{\mathbf{d}\,\Delta \boldsymbol{k}}{2}(x - x_0) - \frac{\Delta \boldsymbol{k}}{2}\mathbf{d}\,x_0\right) \\
&+\mathbf{d}P \times F \times P^{-1} \times \Delta \boldsymbol{k} + P \times dF_1 \times P^{-1} \times \Delta \boldsymbol{k}\,\mathbf{d}\,x_0 + P \times \left(dF_2 \odot \mathbf{d}\,(diag\,\boldsymbol{\beta})\right) \times P^{-1} \times \Delta \boldsymbol{k} \\
&- P \times F \times P^{-1} \times \mathbf{d}\,P \times P^{-1} \times \Delta \boldsymbol{k} \\
&+ P \times F \times P^{-1} \times \mathbf{d}\Delta \boldsymbol{k} \qquad (20)
\end{aligned}
$$

Take the vectorization of each term of Eq. (13),

$$vec\left(\mathbf{d}\,P \times E \times P^{-1} \times \left(\frac{\Delta \boldsymbol{w}}{2} + \frac{\Delta \boldsymbol{k}}{2}(x - x_0)\right)\right)$$
$$= \left(\left(E \times P^{-1} \times \left(\frac{\Delta \boldsymbol{w}}{2} + \frac{\Delta \boldsymbol{k}}{2}(x - x_0)\right)\right)^{\mathrm{T}} \otimes I_n\right) \times vec(\mathbf{d}\,P) \tag{21}$$

$$vec\left(P \times \left(dE_2 \odot \mathbf{d}\,(diag\,\boldsymbol{\beta})\right) \times P^{-1} \times \left(\frac{\Delta \boldsymbol{w}}{2} + \frac{\Delta \boldsymbol{k}}{2}(x - x_0)\right)\right)$$
$$= \left(\left(P^{-1} \times \left(\frac{\Delta \boldsymbol{w}}{2} + \frac{\Delta \boldsymbol{k}}{2}(x - x_0)\right)\right)^{T} \otimes P\right) \times vec\left(\left(dE_2 \odot \mathbf{d}\,(diag\,\boldsymbol{\beta})\right)\right)$$
$$= \left(\left(P^{-1} \times \left(\frac{\Delta \boldsymbol{w}}{2} + \frac{\Delta \boldsymbol{k}}{2}(x - x_0)\right)\right)^{T} \otimes P\right) \times diag\left(vec(dE_2)\right) \times vec\left(\mathbf{d}\,(diag\,\boldsymbol{\beta})\right) \tag{22}$$

$$vec\left(-P \times E \times P^{-1} \times \mathbf{d}\,P \times P^{-1} \times \left(\frac{\Delta \boldsymbol{w}}{2} + \frac{\Delta \boldsymbol{k}}{2}(x - x_0)\right)\right)$$
$$= -\left(\left(P^{-1} \times \left(\frac{\Delta \boldsymbol{w}}{2} + \frac{\Delta \boldsymbol{k}}{2}(x - x_0)\right)\right)^{T} \otimes (P \times E \times P^{-1})\right) \times vec(\mathbf{d}\,P) \tag{23}$$

$$vec(\mathbf{d}P \times F \times P^{-1} \times \Delta \boldsymbol{k})$$
$$= \left((F \times P^{-1} \times \Delta \boldsymbol{k})^{\mathrm{T}} \otimes I_n\right) \times vec(\mathbf{d}\,P) \tag{24}$$

$$vec\left(P \times \left(dF_2 \odot \mathbf{d}\,(diag\,\boldsymbol{\beta})\right) \times P^{-1} \times \Delta \boldsymbol{k}\right)$$
$$= ((P^{-1} \times \Delta \boldsymbol{k})^{T} \otimes P) \times vec\left(\left(dF_2 \odot \mathbf{d}\,(diag\,\boldsymbol{\beta})\right)\right)$$
$$= ((P^{-1} \times \Delta \boldsymbol{k})^{T} \otimes P) \times diag\left(vec(dF_2)\right) \times vec\left(\mathbf{d}\,(diag\,\boldsymbol{\beta})\right) \tag{25}$$

$$vec(-P \times F \times P^{-1} \times \mathbf{d}\,P \times P^{-1} \times \Delta \boldsymbol{k})$$
$$= -\left((P^{-1} \times \Delta \boldsymbol{k})^{T} \otimes (P \times F \times P^{-1})\right) \times vec(\mathbf{d}\,P) \tag{26}$$

Therefore, the vectorization of $\mathbf{d}\,\boldsymbol{w}$ is

$$vec(\mathbf{d}\,\boldsymbol{w})$$

$$= \mathbf{d}\,\overline{\boldsymbol{w}} + P \times E \times P^{-1} \times \frac{\mathbf{d}\,\Delta\boldsymbol{w}}{2} + (x - x_0)\mathbf{d}\,\overline{\boldsymbol{k}} + \left(\left(\frac{x - x_0}{2}\right) P \times E \times P^{-1} + \; P \times F \times P^{-1}\right) \times \mathbf{d}\,\Delta\boldsymbol{k}$$

$$\left\{\left(\left(E \times P^{-1} \times \left(\frac{\Delta\boldsymbol{w}}{2} + \frac{\Delta\boldsymbol{k}}{2}(x - x_0)\right)\right)^{\mathrm{T}} \otimes I_n\right) - \left(\left(P^{-1} \times \left(\frac{\Delta\boldsymbol{w}}{2} + \frac{\Delta\boldsymbol{k}}{2}(x - x_0)\right)\right)^{T} \otimes (P \times E \times P^{-1})\right)\right.$$

$$\left. + \left((F \times P^{-1} \times \Delta\boldsymbol{k})^{\mathrm{T}} \otimes I_n\right) - \left((P^{-1} \times \Delta\boldsymbol{k})^{T} \otimes (P \times F \times P^{-1})\right)\right\} \times vec(\mathrm{d}P)$$

$$+ \left\{\left(\left(P^{-1} \times \left(\tfrac{\Delta\boldsymbol{w}}{2} + \tfrac{\Delta\boldsymbol{k}}{2}(x - x_0)\right)\right)^{T} \otimes P\right) \times diag\big(vec(dE_2)\big) + ((P^{-1} \times \Delta\boldsymbol{k})^{T} \otimes P) \times\right.$$

$$\left. diag\big(vec(dF_2)\big)\right\} \times vec\big(\mathbf{d}\,(diag\,\boldsymbol{\beta})\big)$$

$$+ \left\{-\overline{\boldsymbol{k}} + P \times dE_1 \times P^{-1} \times \left(\tfrac{\Delta\boldsymbol{w}}{2} + \tfrac{\Delta\boldsymbol{k}}{2}(x - x_0)\right) - P \times E \times P^{-1} \times \tfrac{\Delta\boldsymbol{k}}{2} + P \times dF_1 \times P^{-1} \times \Delta\boldsymbol{k}\right\} \mathbf{d}\,x_0$$

(27)

Therefore,

$$w_{x_0} = \left(-\overline{\boldsymbol{k}} + P \times dE_1 \times P^{-1} \times \left(\tfrac{\Delta\boldsymbol{w}}{2} + \tfrac{\Delta\boldsymbol{k}}{2}(x - x_0)\right) - P \times E \times P^{-1} \times \tfrac{\Delta\boldsymbol{k}}{2} + P \times dF_1 \times P^{-1} \times \Delta\boldsymbol{k}\right)^{T} \quad (28)$$

$$w_{\overline{\boldsymbol{w}}} = \frac{\partial w}{\partial \overline{\boldsymbol{w}}} = I_n \quad (29)$$

$$w_{\Delta\boldsymbol{w}} = \frac{\partial w}{\partial \Delta\boldsymbol{w}} = \left\{\frac{P \times E \times P^{-1}}{2}\right\}^{T} \quad (30)$$

$$w_{\overline{\boldsymbol{k}}} = \frac{\partial w}{\partial \overline{\boldsymbol{k}}} = (x - x_0) I_n \quad (31)$$

$$w_{\Delta\boldsymbol{k}} = \frac{\partial w}{\partial \Delta\boldsymbol{k}} = \left\{\left(\frac{x - x_0}{2}\right) P \times E \times P^{-1} + \; P \times F \times P^{-1}\right\}^{T} \quad (32)$$

$$w_P = \frac{\partial w}{\partial P} = \left\{ \left(\left(E \times P^{-1} \times \left(\frac{\Delta \boldsymbol{w}}{2} + \frac{\Delta \boldsymbol{k}}{2}(x - x_0) \right) \right)^{\mathrm{T}} \otimes I_n \right) - \left(\left(P^{-1} \times \left(\frac{\Delta \boldsymbol{w}}{2} + \frac{\Delta \boldsymbol{k}}{2}(x - x_0) \right) \right)^{T} \otimes (P \times E \times P^{-1}) \right) + \left((F \times P^{-1} \times \Delta \boldsymbol{k})^{\mathrm{T}} \otimes I_n \right) - \left((P^{-1} \times \Delta \boldsymbol{k})^{T} \otimes (P \times F \times P^{-1}) \right) \right\}^{T}$$

$$= \left(\left(E \times P^{-1} \times \left(\frac{\Delta \boldsymbol{w}}{2} + \frac{\Delta \boldsymbol{k}}{2}(x - x_0) \right) \right) \quad \otimes I_n \right) - \left(\left(P^{-1} \times \left(\frac{\Delta \boldsymbol{w}}{2} + \frac{\Delta \boldsymbol{k}}{2}(x - x_0) \right) \right) \quad \otimes (P \times E \times P^{-1})^{T} \right) + \left((F \times P^{-1} \times \Delta \boldsymbol{k}) \quad \otimes I_n \right) - \left((P^{-1} \times \Delta \boldsymbol{k}) \quad \otimes (P \times F \times P^{-1})^{T} \right) \tag{33}$$

and

$$w_{diag\ \boldsymbol{\beta}} = \frac{\partial w}{\partial diag\ \boldsymbol{\beta}} = \left\{ \left(\left(P^{-1} \times \left(\frac{\Delta \boldsymbol{w}}{2} + \frac{\Delta \boldsymbol{k}}{2}(x - x_0) \right) \right)^{T} \otimes P \right) \times diag\left(vec(dE_2)\right) + ((P^{-1} \times \Delta \boldsymbol{k})^{T} \otimes P) \times diag\left(vec(dF_2)\right) \right\}^{T} \tag{34}$$

To get $\frac{\partial w_{n+1}}{\partial x_0}$, $\frac{\partial w_{n+1}}{\partial \overline{\boldsymbol{w}}}$, $\frac{\partial w_{n+1}}{\partial \Delta \boldsymbol{w}}$, $\frac{\partial w_{n+1}}{\partial \overline{\boldsymbol{k}}}$, $\frac{\partial w_{n+1}}{\partial \Delta \boldsymbol{k}}$, $\frac{\partial w_{n+1}}{\partial P}$, and $\frac{\partial w_{n+1}}{\partial diag\ \boldsymbol{\beta}}$:

$$\frac{\partial w_{n+1}}{\partial x_0} = -\sum_i^n w_{x_0}(:, i) \tag{35}$$

$$\frac{\partial w_{n+1}}{\partial \overline{\boldsymbol{w}}} = -\sum_i^n w_{\overline{\boldsymbol{w}}}(:, i) \tag{36}$$

$$\frac{\partial w_{n+1}}{\partial \Delta \boldsymbol{w}} = -\sum_i^n w_{\Delta \boldsymbol{w}}(:, i) \tag{37}$$

$$\frac{\partial w_{n+1}}{\partial \overline{\boldsymbol{k}}} = -\sum_i^n w_{\overline{\boldsymbol{k}}}(:, i) \tag{38}$$

$$\frac{\partial w_{n+1}}{\partial \Delta \boldsymbol{k}} = -\sum_i^n w_{\Delta \boldsymbol{k}}(:, i) \tag{39}$$

$$\frac{\partial w_{n+1}}{\partial P} = -\sum_i^n w_P(:, i) \tag{40}$$

$$\frac{\partial w_{n+1}}{\partial diag\ \boldsymbol{\beta}} = -\sum_i^n w_{diag\ \boldsymbol{\beta}}(:, i) \tag{41}$$

Since $\boldsymbol{r} = {}^{(\boldsymbol{w_0} - \boldsymbol{w}\quad)}./_{\boldsymbol{\sigma}}$, we have

$$r_{x_0} = {}^{-w_{x_0}\cdot}\!\big/_{\sigma} \tag{42}$$

$$r_{\bar{\boldsymbol{w}}} = {}^{-w_{\bar{\boldsymbol{w}}}\,\cdot}\!\big/_{\sigma} \tag{43}$$

$$r_{\Delta\boldsymbol{w}} = {}^{-w_{\Delta\boldsymbol{w}}\cdot}\!\big/_{\sigma} \tag{44}$$

$$r_{\bar{\boldsymbol{k}}} = {}^{-w_{\bar{\boldsymbol{k}}}\,\cdot}\!\big/_{\sigma} \tag{45}$$

$$r_{\Delta\boldsymbol{k}} = {}^{-w_{\Delta\boldsymbol{k}}\cdot}\!\big/_{\sigma} \tag{46}$$

$$r_{P} = {}^{-w_{P}\,\cdot}\!\big/_{\sigma} \tag{47}$$

$$r_{diag\,\boldsymbol{\beta}} = {}^{-w_{diag\,\boldsymbol{\beta}}\cdot}\!\big/_{\sigma} \tag{48}$$

To get $\frac{\partial r_{n+1}}{\partial x_0}$, $\frac{\partial r_{n+1}}{\partial \bar{\boldsymbol{w}}}$, $\frac{\partial r_{n+1}}{\partial \Delta\boldsymbol{w}}$, $\frac{\partial r_{n+1}}{\partial \bar{\boldsymbol{k}}}$, $\frac{\partial r_{n+1}}{\partial \Delta\boldsymbol{k}}$, $\frac{\partial r_{n+1}}{\partial P}$, and $\frac{\partial r_{n+1}}{\partial diag\,\boldsymbol{\beta}}$:

$$\frac{\partial r_{n+1}}{\partial x_0} = {}^{\left(\sum_i^n w_{x_0}\,(:,i)\right)}\!\Big/_{\sigma_{n+1}} \tag{49}$$

$$\frac{\partial r_{n+1}}{\partial \bar{\boldsymbol{w}}} = {}^{\left(\sum_i^n w_{\bar{\boldsymbol{w}}}\,(:,i)\right)}\!\Big/_{\sigma_{n+1}} \tag{50}$$

$$\frac{\partial r_{n+1}}{\partial \Delta\boldsymbol{w}} = {}^{\left(\sum_i^n w_{\Delta\boldsymbol{w}}\,(:,i)\right)}\!\Big/_{\sigma_{n+1}} \tag{51}$$

$$\frac{\partial r_{n+1}}{\partial \bar{\boldsymbol{k}}} = {}^{\left(\sum_i^n w_{\bar{\boldsymbol{k}}}\,(:,i)\right)}\!\Big/_{\sigma_{n+1}} \tag{52}$$

$$\frac{\partial r_{n+1}}{\partial \Delta\boldsymbol{k}} = {}^{\left(\sum_i^n w_{\Delta\boldsymbol{k}}\,(:,i)\right)}\!\Big/_{\sigma_{n+1}} \tag{53}$$

$$\frac{\partial r_{n+1}}{\partial P} = {}^{\left(\sum_i^n w_{P}\,(:,i)\right)}\!\Big/_{\sigma_{n+1}} \tag{54}$$

$$\frac{\partial r_{n+1}}{\partial diag\,\boldsymbol{\beta}} = {}^{\left(\sum_i^n w_{diag\,\boldsymbol{\beta}}\,(:,i)\right)}\!\Big/_{\sigma_{n+1}} \tag{55}$$